JOINT SESSION · WEEK 3

Data workshop

Comparing Europe with Data. Be able to load and document the COIL data in your tool (Python, R, Gretl or Power BI) before the analysis starts.

Week 3 · 2 Nov – 8 Nov 2026 · 45 minutes · Microsoft Teams

Ulysseus EUROPEAN UNIVERSITY

UNIVERSIDAD DE SEVILLA

TECHNICAL UNIVERSITY OF KOŠICE

UCG Univerzitet Crne Gore

AGENDA · 45 MINUTES

What we will do today



5 min	The dataset: indicators, sources, coverage and gaps (especially Montenegro).
20 min	Live demo, about 5 minutes per tool: loading the data, selecting countries and indicators and a first chart in Python, R, Gretl and Power BI.
5 min	How to write the data dictionary and handle missing values.
15 min	Questions from the teams.

THE DATASET

One row per country and year

iso3	country	year	SL_UEM_1524_ZS	NY_GDP_PCAP_PP_KD	IT_NET_USER_ZS
ESP	Spain	2022	29.7	46733	94.5
ESP	Spain	2023	28.7	47323	95.4
SVK	Slovakia	2022	19.9	38649	89.1
SVK	Slovakia	2023	19.9	39502	87.2
MNE	Montenegro	2022	29.8	25583	88.2
MNE	Montenegro	2023	23.3	27243	88.6

Cross-section = one year, many countries. Panel = many countries followed over many years. Codes are World Bank / IMF codes; in this file the dots are replaced by underscores, so the same names work in Python, R, Gretl and Power BI.

Before any analysis, drop the row "EUU" (EU aggregate): it is not a country.

DATA SECTION OF THE WEBSITE

Three files with the same data

coil_indicators_gretl.csv

One row per country and year; codes with underscores. Use it in Gretl, Python and R (all the code in the methods module uses this file).

coil_indicators_wide.csv

One row per country and year; original codes with dots. Easiest for Excel and Power BI.

coil_indicators_long.csv

One row per country, year and indicator, with unit and source in every row. Handy for the data dictionary.

The Data section also shows the coverage of every indicator by country, and the list of indicators with their units.

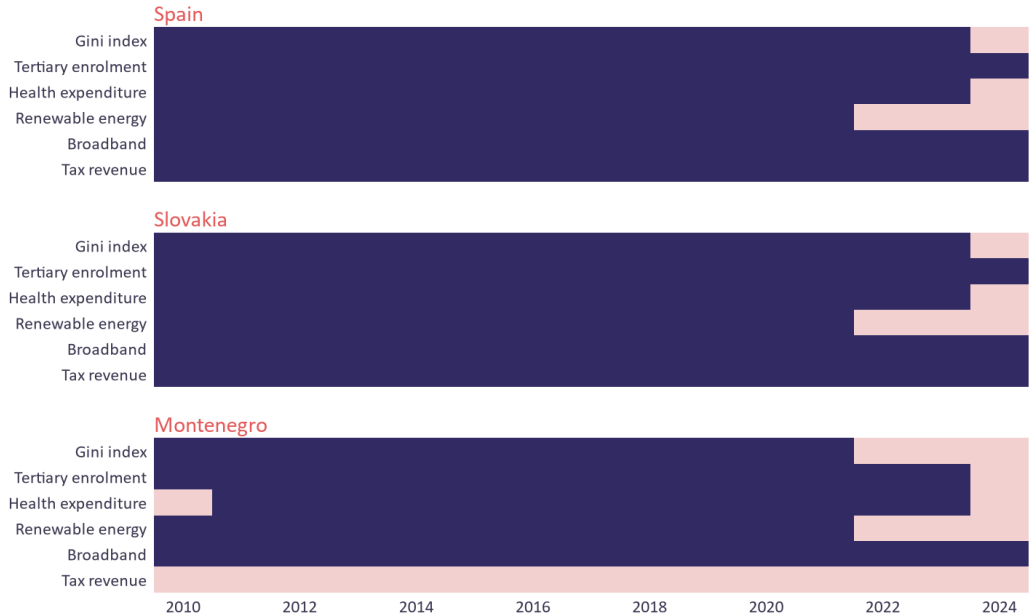
21 INDICATORS

What is in the dataset

Theme	Indicators (unit)
Labour market	Unemployment, total (% of labour force (modelled ILO estimate)) · Unemployment, youth (15-24) (% of labour force aged 15-24) · Female labour force participation (% of female population 15+)
Income and growth	GDP per capita, PPP (constant 2021 international $) · GDP growth (annual %) · Inflation, consumer prices (annual %) · Services, value added (% of GDP)
Inequality	Gini index (index (0-100))
Public finance	General government gross debt (% of GDP) · Tax revenue (% of GDP)
Demography	Population, total (persons) · Population aged 65 and above (% of total population) · Fertility rate (births per woman)
Education	Tertiary school enrolment (% gross) · Government expenditure on education (% of GDP)
Health	Life expectancy at birth (years) · Current health expenditure (% of GDP)
Digitalisation	Individuals using the Internet (% of population) · Fixed broadband subscriptions (per 100 people)
Environment	Renewable energy consumption (% of total final energy consumption) · CO2 emissions per capita (excl. LULUCF) (t CO2e per capita)

COVERAGE

Gaps you have to know about



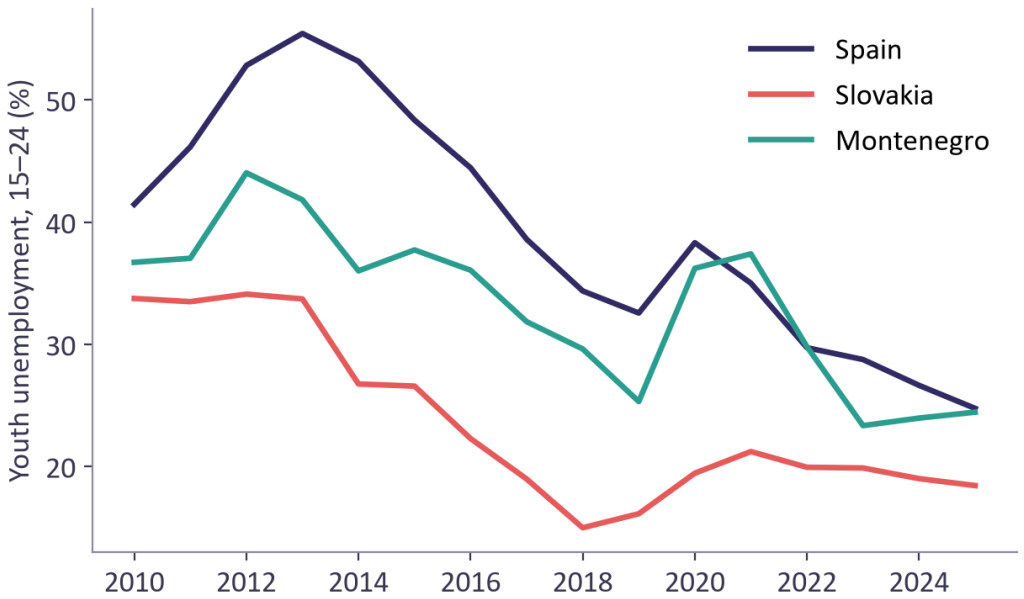
Dark = data available, pink = missing (2010–2024)

What this means

- Montenegro has no tax revenue data (15 of 15 years missing), and no CO2 or education spending series.
- Renewable energy ends in 2021 for every country; the Gini index ends in 2022–2023.
- Health expenditure ends in 2023.
- Check coverage before you choose your indicators and years.

LIVE DEMO · 20 MINUTES

The same first chart in four tools



The task

- Load the COIL file.
- Keep Spain, Slovakia and Montenegro.
- Plot youth unemployment (SL_UEM_1524_ZS) from 2010 to 2025.
- Each tool takes about 5 minutes: first without code (Power BI, Gretl), then with code (Python, R).

DEMO · NO CODE NEEDED

Without writing code: Power BI and Gretl

Power BI Desktop

- Get data > Text/CSV > coil_indicators_wide.csv > Load.
- Insert a Line chart.
- X-axis: year. Y-axis: SL.UEM.1524.ZS. Legend: country.
- Filters pane: country is Spain, Slovakia, Montenegro.
- Check that year is a whole number, not summarised.

Gretl

- File > Open data > User file > coil_indicators_gretl.csv.
- Data > Dataset structure > Panel: unit = iso3, time = year.
- Sample > Restrict, based on criterion: iso3 == "ESP" || iso3 == "SVK" || iso3 == "MNE".
- For the chart, type one line in the Gretl console: panplot SL_UEM_1524_ZS --overlay

Both routes give the chart on the previous slide. The step-by-step guide for each method is in the methods module.

DEMO · OPTIONAL

With code: Python and R (if your course uses them)

Python (pandas + matplotlib)

```
import pandas as pd
import matplotlib.pyplot as plt

d = pd.read_csv("coil_indicators_gretl.csv")
h = d[d.iso3.isin(["ESP", "SVK", "MNE"])]
h.pivot(index="year", columns="country",
        values="SL_UEM_1524_ZS").plot()
plt.ylabel("Youth unemployment, 15-24 (%)")
plt.show()
```

R (ggplot2)

```
d <- read.csv("coil_indicators_gretl.csv")
h <- subset(d, iso3 %in% c("ESP", "SVK", "MNE"))

library(ggplot2)
ggplot(h, aes(year, SL_UEM_1524_ZS,
              colour = country)) +
  geom_line() +
  labs(y = "Youth unemployment, 15-24 (%)")
```

Run the code from the folder where you saved the CSV file. In R, install ggplot2 once with install.packages("ggplot2").

PART OF DELIVERABLE 2

The data dictionary: one row per variable

Variable	Code	Definition	Unit	Source	Years	Missing values and what we did
youth_unemp	SL.UEM.1524.ZS	Unemployment, youth (15–24)	% of labour force 15–24	World Bank	2010–2025	None
health_exp	SH.XPD.CHEX.GD.ZS	Current health expenditure	% of GDP	World Bank	2010–2023	Montenegro 2010 missing: Montenegro enters in 2011 (unbalanced panel)
tax	GC.TAX.TOTL.GD.ZS	Tax revenue	% of GDP	World Bank	2010–2024	No data for Montenegro: country left out of this variable, stated in the report

Write it as soon as you choose your indicators. It saves time in every later step and it is part of deliverable 2.

MISSING VALUES

Never fill a gap silently

1. Change the year or the indicator

Use a year or an indicator with full coverage for your countries.

2. Leave the country out

For that variable only, and say so in the report and in every table (n = …).

3. Interpolate inside a country

Only between two known years, never beyond the last one. Mark the values you created.

4. Add another source

Eurostat or the national statistics office (e.g. MONSTAT). Document it in the data dictionary.

BY THE END OF WEEK 3

Your checklist

- [ ] Your dataset is ready (one row per country and year) and every variable is documented in a data dictionary.
- [ ] Every variable has its code, unit, source and years in the data dictionary.
- [ ] You know which gaps affect your countries and what you will do with them.
- [ ] Week 4: exploratory data analysis. Descriptive statistics and first charts are done, and the team agrees on what the data show.

Questions from the teams · 15 minutes